## PL/ SQL EXERCISES

### Assumptions

Since the database schema was not provided in the assignment, the following tables were created to implement and test the PL/SQL solutions.

### Customers Table

```
CREATE TABLE Customers (
   CustomerID NUMBER,
   Name VARCHAR2(20),
   Age NUMBER,
   Balance NUMBER,
   IsVIP VARCHAR2(5)
);
```

### Loans Table

```
CREATE TABLE Loans (
   LoanID NUMBER,
   CustomerID NUMBER,
   InterestRate NUMBER,
   DueDate DATE
);
```

### Sample Data:

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.185 seconds

| | CUSTOMERID | NAME | AGE | BALANCE | ISVIP |
|---|---|---|---|---|---|
| 1 | 1 | John | 65 | 15000 | NO |
| 2 | 2 | Alice | 45 | 8000 | NO |
| 3 | 3 | Bob | 70 | 12000 | NO |
| 4 | 4 | David | 55 | 5000 | NO |
| 5 | 5 | Emma | 62 | 18000 | NO |
| 6 | 6 | Frank | 35 | 7000 | NO |
| 7 | 7 | Grace | 68 | 25000 | NO |
| 8 | 8 | Henry | 50 | 9500 | NO |
| 9 | 9 | Ivy | 72 | 30000 | NO |
| 10 | 10 | Jack | 40 | 11000 | NO |

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.162 seconds

| | LOANID | CUSTOMERID | INTERESTRATE | DUEDATE |
|---|---|---|---|---|
| 1 | 101 | 1 | 8 | 7/10/2026, 9:47:55 |
| 2 | 102 | 2 | 10 | 7/30/2026, 9:47:55 |
| 3 | 103 | 3 | 9 | 7/5/2026, 9:47:55 A |
| 4 | 104 | 4 | 11 | 8/9/2026, 9:47:55 A |
| 5 | 105 | 5 | 7 | 6/30/2026, 9:47:55 |
| 6 | 106 | 6 | 12 | 8/19/2026, 9:47:55 |
| 7 | 107 | 7 | 8 | 7/15/2026, 9:47:55 |
| 8 | 108 | 8 | 9 | 7/25/2026, 9:47:55 |
| 9 | 109 | 9 | 6 | 6/25/2026, 9:47:55 |
| 10 | 110 | 10 | 10 | 7/8/2026, 9:47:55 A |

**Exercise 1: Control Structures**

**Scenario 1:** The bank wants to apply a discount to loan interest rates for customers above 60 years old.

- **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

  **PL/SQL Code:**

```
BEGIN
  FOR c IN (
    SELECT CustomerID, Age
    FROM Customers
  )
  LOOP
    IF c.Age > 60 THEN
      UPDATE Loans
      SET InterestRate = InterestRate - 1
      WHERE CustomerID = c.CustomerID;
    END IF;
  END LOOP;

  COMMIT;
END;
/
```

  **OUTPUT VERIFICATION:**

SELECT * FROM Loans;

**OUTPUT:**

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.007 seconds

| | LOANID | CUSTOMERID | INTERESTRATE | DUEDATE |
|---|---|---|---|---|
| 1 | 101 | 1 | 7 | 7/10/2026, 9:47:55 |
| 2 | 102 | 2 | 10 | 7/30/2026, 9:47:55 |
| 3 | 103 | 3 | 8 | 7/5/2026, 9:47:55 A |
| 4 | 104 | 4 | 11 | 8/9/2026, 9:47:55 A |
| 5 | 105 | 5 | 6 | 6/30/2026, 9:47:55 |
| 6 | 106 | 6 | 12 | 8/19/2026, 9:47:55 |
| 7 | 107 | 7 | 7 | 7/15/2026, 9:47:55 |
| 8 | 108 | 8 | 9 | 7/25/2026, 9:47:55 |
| 9 | 109 | 9 | 5 | 6/25/2026, 9:47:55 |
| 10 | 110 | 10 | 10 | 7/8/2026, 9:47:55 A |

**Scenario 2: A customer can be promoted to VIP status based on their balance.**

- **Question: Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over $10,000.**

**PL/SQL CODE:**

```
BEGIN
  FOR c IN (
    SELECT CustomerID, Balance
    FROM Customers
  )
  LOOP
    IF c.Balance > 10000 THEN
      UPDATE Customers
      SET IsVIP = 'YES'
      WHERE CustomerID = c.CustomerID;
    END IF;
  END LOOP;

  COMMIT;
END;
/
```

**OUTPUT VERIFICATION:**

SELECT * FROM CUSTOMERS;

**OUTPUT:**

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.007 seconds

| | CUSTOMERID | NAME | AGE | BALANCE | ISVIP |
|---|---|---|---|---|---|
| 1 | 1 | John | 65 | 15000 | YES |
| 2 | 2 | Alice | 45 | 8000 | NO |
| 3 | 3 | Bob | 70 | 12000 | YES |
| 4 | 4 | David | 55 | 5000 | NO |
| 5 | 5 | Emma | 62 | 18000 | YES |
| 6 | 6 | Frank | 35 | 7000 | NO |
| 7 | 7 | Grace | 68 | 25000 | YES |
| 8 | 8 | Henry | 50 | 9500 | NO |
| 9 | 9 | Ivy | 72 | 30000 | YES |
| 10 | 10 | Jack | 40 | 11000 | YES |

**Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.**

- **Question: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.**

**PL/SQL CODE:**

```
BEGIN
  FOR l IN (
    SELECT CustomerID, DueDate
    FROM Loans
    WHERE DueDate <= SYSDATE + 30
  )
  LOOP
    DBMS_OUTPUT.PUT_LINE(
      'Reminder for Customer '
      || l.CustomerID
      || ' Loan Due On '
      || l.DueDate
    );
  END LOOP;
```

```
END;
/
```

**OUTPUT:**

```
Reminder for Customer 1 Loan Due On 10-JUL-26
Reminder for Customer 3 Loan Due On 05-JUL-26
Reminder for Customer 5 Loan Due On 30-JUN-26
Reminder for Customer 7 Loan Due On 15-JUL-26
Reminder for Customer 9 Loan Due On 25-JUN-26
Reminder for Customer 10 Loan Due On 08-JUL-26


PL/SQL procedure successfully completed.
```